

RYAN JAMES F. INDANGAN

AI Software Engineer • Full-Stack Developer • Certified CTO

RÉSUMÉ

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UpWork • LinkedIn • GitHub Portfolio

PROFESSIONAL SUMMARY

AI Software Engineer and full-stack developer with 8+ years building production systems end to end — from AI/ML and LLM-powered automation to secure, scalable web platforms. Specializes in **health technology** — registry-scale trial data engineering, clinical-trial business development, and AI research intelligence — alongside document intelligence (OCR, computer vision, ML-based extraction) and LLM integration, with delivery across clinical research, financial underwriting automation, and AI companion applications. A Certified CTO and startup co-founder who has led cross-functional teams of up to 12 engineers and consistently turns ambiguous requirements into shipped, reliable software.

CORE STRENGTHS

Health Technology & Clinical Research

- Clinical-trial intelligence platforms: study/site/investigator entity modelling, registry ingestion and normalisation, provenance tracking, entity resolution and deduplication
- Clinical-trial business development: pipeline tracking, opportunity scoring, registry-driven lead discovery
- ClinicalTrials.gov integration — live status synchronisation, phase-change detection, registry reconciliation
- Audit & integrity engineering: append-only SHA-256 hash-chained records with scheduled chain verification, role-based access control, TOTP 2FA, reversible data layers
- AI-compiled background research — cited, confidence-rated briefs over trial and market signals, stored as an additive layer that never mutates source records

AI & Machine Learning

- Document intelligence & OCR (Tesseract, OpenCV, PyMuPDF, pdfplumber); computer vision and image processing
- Machine learning (scikit-learn, Random Forests, feature engineering); ML-based watermark detection & removal
- LLM integration & orchestration (Ollama, OpenAI-compatible APIs, vLLM); conversational AI & companion apps
- End-to-end AI pipelines with real-time streaming (FastAPI SSE) and validation; model deployment (Hugging Face)

Web, API & Real-Time Systems

- RESTful & GraphQL APIs, WebSockets, Server-Sent Events; real-time and transaction-processing systems
- Full-stack development: React, Next.js, Vue, Angular, Node.js, TypeScript, PHP, Python
- Secure backend services: JWT/RSA auth, bcrypt, rate limiting, CORS, hardened API gateways

Workflow Automation & Integration

- Process & workflow automation (n8n), CI/CD automation, data transformation pipelines
- Legacy-to-custom migrations (e.g., Salesforce → bespoke web apps); REST/third-party integrations; Excel reporting (openpyxl)

Frameworks & Libraries

- FastAPI, Express.js, Laravel, CodeIgniter 3, Django
- OpenCV, scikit-image, Pillow, Tesseract OCR, pdfplumber, pdf2image, PyMuPDF

Cloud, DevOps & Databases

- AWS (EC2, RDS, Lambda, EventBridge, S3, CloudFront), DigitalOcean, Docker, Docker Compose, Kubernetes, CI/CD
- MySQL, PostgreSQL, SQLite, MongoDB, Redis, Firebase; data normalization and structured extraction

Programming & Scripting

- Python (AI/ML pipelines, PDF & image processing), JavaScript/TypeScript, PHP, SQL, HTML/CSS, Bash

Tools, Testing & Project Management

- Git, GitHub/GitLab, Postman, Insomnia; Pytest, Jest, Mocha, Selenium, Cypress
- Jira, Confluence, Asana, Trello, Slack, MS Teams, ClickUp, Monday.com

WORK EXPERIENCE

HireOverseas (deployed to Core Clinical Trials) — AI Workflow Engineer

May 2026 – Present

Remote **Independent Contractor**

- Engaged through HireOverseas and deployed to Core Clinical Trials as lead developer across four production systems — two internal clinical-trial intelligence platforms and two public-facing sites with custom CMS engines.
- **Clinical Trial Intelligence & BD Platform (internal name: CDMS)** — architected and shipped a custom trial-intelligence and business-development platform to AWS (Elastic Beanstalk, RDS), moving the organization's study tracker and BD pipeline off Salesforce to a tailored web application that eliminates licensing overhead and matches their exact workflows. Models trials, sites, investigators, and sponsors keyed on the ClinicalTrials.gov NCT ID rather than a generic CRM object, with a tamper-evident SHA-256 hash-chained audit trail whose models reject UPDATE at the ORM layer, role-based access, and TOTP two-factor authentication.
- **BD Pipeline** (within the CDMS) — built the clinical-trial business-development engine: spreadsheet ingestion with defensive parsing and non-destructive idempotent upserts, live ClinicalTrials.gov status sync that flags registry status changes, SEC-filing and news lead discovery, a shared lead-tracking CRM with owner-aware follow-up reminders, and AI-assisted outreach drafting with cross-source duplicate-contact guards.
- **Core Research Intelligence** (standalone platform — own repo, composer package, ~27-table schema and admin panel; later integrated into the CDMS as an isolated add-on with the host schema untouched) — built a resumable, checkpointed ClinicalTrials.gov ETL with entity resolution across studies, sites, investigators, and sponsors, explainable rules-based site classification, a 0-100 opportunity score with runtime-editable weights, and human-reviewed, reversible deduplication over a 150,000-site corpus.
- **AI research enrichment** (within the CDMS) — designed a reversible enrichment layer that generates cited, confidence-rated background briefs on priority trials and market signals, recording explicitly what could not be verified; additive by design so source records are never mutated, and clearable by batch, type, or record.
- **TrialPulse** — delivered a marketing and demo platform with a custom CMS engine (full edit history, one-click revert, sandboxed preview) so non-technical staff can safely edit the live site themselves.
- **Corporate site rework** — rebuilt the organization's public website from the ground up on a new enterprise design system, staged behind a private preview until vetted.
- Automating labor-intensive manual processes for the CEO and Executive Director, reducing operational workload and turnaround time.

FirstKind AI — AI Software Engineer & Co-Founder, GO Models

Jan 2026 – Present

Remote **Independent Contractor**

- Co-founded GO Models, a suite of AI companion applications, serving as founding AI Software Engineer.
- Architect and build LLM-powered conversational experiences end-to-end — model integration, backend APIs, and frontend.
- Drive product and technical direction from prototype to deployment, establishing engineering standards and AI infrastructure.

Alliance Global Solutions BPO Intl Corp. — AI Developer / Machine Learning Engineer

Nov 2025 – Apr 2026

Alternative Funding Group · On-site — SM North EDSA, Tower 1, Quezon City **Full-time**

- Built an end-to-end bank statement extraction and validation pipeline combining native PDF text extraction with OCR fallback to support automated underwriting workflows.
- Implemented advanced OCR preprocessing using OpenCV and PIL — including denoising, adaptive thresholding, and deskewing — with aggressive multi-angle OCR for difficult scans selected via quality scoring.
- Designed a watermark-removal ML subsystem using feature engineering and RandomForest classification to auto-detect watermark types and route documents to appropriate removal strategies.
- Created a validation-first extraction framework enforcing structural integrity invariants and auditability without mutating source financial text.
- Developed deterministic wire and revenue classification with evidence-backed rules to support compliance and underwriting review.
- Integrated local LLM workflows via Ollama with OpenAI-compatible backends to generate structured underwriting summaries with token-efficient payloads.
- Implemented a secure service architecture with JWT authentication, RSA key management, and a hardened Node gateway using rate limiting, CORS allowlists, and bcrypt-based authentication.
- Delivered real-time processing visibility via FastAPI Server-Sent Events to stream AI pipeline phases and results to the frontend.

Bada LLC — Support Engineer

Apr 2025 – Oct 2025

Remote

- Developed and scripted solutions for data management and client image migrations, improving operational efficiency.
- Created custom templates for InDesign, Word, and PowerPoint to standardize client deliverables.
- Integrated systems using REST APIs and JavaScript, enabling cross-platform automation.
- Troubleshoot server-side, end-user, and codebase issues, maintaining high client satisfaction.
- Managed client support tickets from initiation to resolution with clear technical guidance.

Skills: JavaScript (ES5), REST APIs, ZenDesk, InDesign, Windows Server, Linux Server, JSON

GlobalX Digital Corporation — Consultant

Jan 2025 – Apr 2025

Hybrid — Pasay City, NCR

- Led the redesign of database architecture, creating scalable, high-performing systems that improved data integrity and business alignment.
- Developed and secured API endpoints, strengthening system communication and overall reliability.
- Directed 12 developers across 3 teams, providing technical guidance, architectural leadership, and workflow documentation.
- Revived a stalled project by assuming ownership, delivering a fully enhanced Version 2 within 2 months after 9 months of failed prior leadership.
- Acted as Technical Lead, aligning goals among cross-functional teams and stakeholders to ensure delivery within strict timelines.

Net Global Solutions Inc. — Back End Developer / Project Lead

Apr 2023 – Nov 2024

Hybrid — Pasig, Metro Manila

- Served as Project Lead & Lead Programmer for the PAGCOR compliance system (Back Office Access), ensuring regulatory adherence.
- Managed the PEZA Digital Marketplace (Builder.ai) as Project Manager, enabling streamlined data management for the Philippine Economic Zone Authority.
- Built a custom PHP framework to enhance API development and system performance.
- Developed a Sports Data Management System, an Automated Billing System, and multiple enterprise solutions as Lead Programmer.

Remote

- Developed the UN-Habitat SDG Cities platform as part of a United Nations initiative.
- Built the Disaster Risk Reduction Management Information System for the Department of Education (DepEd).
- Created a database automated-remapping system ("Bucket Service") to optimize DepEd's database management.

Freelance — Web Developer

2019 – 2023

Remote

- Designed and developed a Sports Data Management System for a private company.
- Built an Online Human Resource Information System for STI Muñoz EDSA.
- Delivered custom websites and applications for small businesses and organizations.

iAm Tech Solutions Inc. — Senior Web Developer

Mar 2018 – Apr 2021

Remote

- Directed large-scale web and mobile projects including a Lotto Play Mobile App, Wholesale E-Commerce, and white-labeled Learning Management Systems.
- Built enterprise systems: Online Inventory & Asset Tracking, Package Tracking, Broker Accreditation (Ayala Land), and Log Monitoring with Digital Signature.
- Delivered custom after-sales service and tracking solutions for corporate clients (Esquire Tech, Chubb Limited, Marmo Pizzara Trading).
- Provided technical leadership as Lead Programmer, ensuring quality delivery and scalable architecture across projects.

SELECTED PROJECTS

Health Tech

- **CDMS — Clinical Trial Intelligence & BD Platform:** AWS-hosted trial-intelligence and BD platform keyed on ClinicalTrials.gov NCT IDs, with a SHA-256 hash-chained tamper-evident audit trail, RBAC and TOTP 2FA.
- **Core Research Intelligence:** Site-intelligence platform with a resumable ClinicalTrials.gov ETL (21,288 studies in 59 min), entity resolution over 150k sites, explainable scoring, and 186 tests on a SQLite+MySQL CI matrix.
- **BD Pipeline & TrialPulse:** A BD engine unifying spreadsheets, registry data and SEC/news signals into a ranked pipeline with a polymorphic lead CRM and AI-assisted outreach; plus a marketing site on a custom CMS engine with edit history and one-click revert.

AI & Machine Learning

- **GO Models:** Co-founded and engineered a suite of LLM-powered AI companion apps end to end as founding AI Software Engineer.
- **Watermark-Removal ML Subsystem:** RandomForest/feature-engineering subsystem that auto-detects watermark types and routes documents to the right removal strategy.

Fin Tech

- **Bank Statement Extraction & Underwriting Pipeline:** End-to-end extraction/underwriting pipeline — native PDF + OCR fallback, OpenCV preprocessing, deterministic classification, local-LLM summaries via Ollama, FastAPI SSE + Node/JWT gateway, React UI.
- **Payment Gateway & Crypto Checkout:** A Django/PostgreSQL payment-gateway backend (merchants, payments, transaction tracking; client-confidential) and an interactive end-to-end crypto checkout flow.

Gaming

- **ARMADO:** Original browser tactical FPS (three.js client + Node/WebSocket server) with a Filament back office featuring a 3D map builder, live weapon tuning and content studios — 100% original code, procedural art and audio.
- **LAAS — Procedural WebGPU World Engine:** Fully procedural in-browser open-world engine (WebGPU) generating every mesh, texture, light and sound from code, with zero bundled assets.
- **iGaming & Betting Platforms:** Multiple real-money gaming backends — casino/lottery, core gaming API, horse-racing and live sabong — with operator back-offices, wallets and crypto payments (client-confidential).

Gov Tech

- **PAGCOR Compliance System:** Led a Symfony/Doctrine regulatory-compliance system for the Philippine gaming regulator PAGCOR (client-confidential).
- **UN-Habitat SDG Cities · DepEd DRRMIS · PEZA Marketplace:** The UN-Habitat SDG Cities platform (United Nations), the Disaster Risk Reduction MIS for the Department of Education, and the PEZA Digital Marketplace (Builder.ai) as Project Manager.

Enterprise, Web & Dev Tools

- **Kayanin & Faith Temple:** A Laravel 13 Philippines-first freelance marketplace with skills-exam qualification, and a Laravel 13 corporate CMS + church-management platform with a fully editable public site.
- **GlobalX, Broker Accreditation (Ayala Land) & more:** Enterprise platforms across real estate, HR (HRIS) and education (LMS).
- **LastResortAI:** A multi-agent dev crew (plan → build → adversarial review → deploy) with persistent memory and multi-pass verification on Claude, Claude Code and the Claude Agent SDK; plus a custom PHP framework and CI/CD automations.

EDUCATION

Bachelor of Science in Information Technology

STI College — Muñoz EDSA

- Relevant Coursework: Data Structures, Algorithms, Database Development, Programming.
- Union Bank UHack 2017 — Semi-Finalist (December 2, 2017).
- HealthHacks PH 2017 — Semi-Finalist (February 17, 2017).
- Battle of the Debuggers 2015 — 1st Runner-up, College (October 2, 2015).

Secondary Education

Graceland Christian College

- Graduated as CAT Corps Commander, leading and coordinating unit activities and drills.
- Member, Performing Arts Guild of Graceland.

CERTIFICATIONS

Leadership & Management

- Certified Chief Technology Officer, Information Technology — Institute of Management, Technology and Finance (IMTF), 2023.

Artificial Intelligence (Anthropic)

- Claude Code in Action — Anthropic (2026). Verify: verify.skilljar.com/c/usugbinq3tyu
- Claude 101 — Anthropic (2026). Verify: verify.skilljar.com/c/ggyq4opwistp

Cybersecurity

- Ethical Hacking Course (Hacker-X) — 21-module program (Sep 2024 – Nov 2025): Know the OS; Hack the Mac; Gathering Information; Leveraging AI for Hacking; Prompt Guide for Hacking; Surfing Anonymously; Hiding Messages; Social Media Hacking; Credit/Debit Card Frauds; Keyboard Spying; WiFi Hacking (WEP, WPA/WPA2); Network Spying; Database Hacking; Android Hacking; Bringing Down a Website; Cross-Site Scripting; Make Your Data Safe; Hacking Passwords; WordPress Scanning; and Vulnerability Scanning & Reporting.

REFERENCES

Available upon request.